

UTLXe on Azure

Deployment and Operations Guide

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Companion guide to *UTL-X: One Language, All Formats*
For UTLXe v1.0 on Azure Container Apps

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1 Quick Start

Deploy UTLXe from the Azure Marketplace, upload your first transformation, and send your first message — all in under 15 minutes.

1.1 What You Get

1.2 Deploy from the Azure Marketplace

1.3 Verify the Deployment

1.4 Upload Your First Transformation

1.5 Test It

1.6 Send a Real Message

1.7 What Just Happened

1.8 Next Steps

2 Writing Transformations

A crash course in UTL-X — just enough to write production transformations. For the complete language reference, see the companion book “UTL-X: One Language, All Formats.”

2.1 The .utlx File Structure

2.2 Property Access

2.3 Object Construction

2.4 Arrays: map, filter, reduce

2.5 Conditionals

2.6 Let Bindings

2.7 Common Standard Library Functions

2.8 Multi-Format Transformations

2.9 User-Defined Functions

2.10 Worked Example: Invoice Transformation

2.11 Where to Learn More

3 The Admin API

Manage transformations, schemas, and bundles via REST on port 8081. This is the primary interface for deploying and operating UTLXe on Azure.

3.1 Architecture: Two Ports, Two Purposes

3.2 Authentication

3.3 Uploading Transformations

3.3.1 Single .utlx File (Simplest)

3.3.2 With Configuration

3.3.3 ZIP Bundle (Batch)

3.4 Managing Schemas

3.5 Testing Before Go-Live

3.6 Listing and Inspecting

3.7 Updating Transformations (Hot-Swap)

3.8 Deleting

3.9 Exporting the Bundle

3.10 Data Plane Discovery

3.11 Engine Info

3.12 Sequence Diagram: Deploy → Test → Go Live

4 Connecting to Azure Services

UTLXe integrates with Azure messaging services via Dapr sidecar. This chapter covers Service Bus, Event Hub, and Event Grid.

4.1 How Dapr Works with UTLXe

4.2 Azure Service Bus

4.2.1 Input: Receiving from a Queue or Topic

4.2.2 Output: Sending to a Queue or Topic

4.2.3 Dead-Letter Queue

4.3 Azure Event Hub

4.3.1 Input: Consuming from Partitions

4.3.2 Output: Publishing Events

4.4 Event Grid (Webhook)

4.5 Worked Example: Dynamics 365 → Peppol

4.6 Dapr Component Configuration Reference

5 Monitoring and Observability

Health probes, Prometheus metrics, and Grafana dashboards for production UTLXe deployments.

5.1 Health Endpoint

5.2 Kubernetes Probes

5.3 Prometheus Metrics

5.4 Setting Up Azure Monitor

5.5 Grafana Dashboard

5.6 Container Logs

5.7 Error Diagnosis

6 Operations

Day-to-day operational tasks: updating transformations, handling incidents, managing validation.

6.1 Updating a Transformation (Zero Downtime)

6.2 Pause and Resume

6.3 Incident Response Workflow

6.4 Validation Override

6.5 Updating Schemas

6.6 Bulk Operations

6.7 Engine Configuration Changes

7 Persistence and Scaling

How transformations survive container restarts, and how to scale UTLXe for throughput.

7.1 The Persistence Problem

7.2 Azure Files Volume Mount (Recommended)

7.3 CI/CD Re-Deploy Pattern

7.4 Startup Sequence

7.5 Memory Sizing

7.6 Scaling

7.7 Never Use Swap

7.8 Poison Messages

8 Security

Network isolation, authentication, and secure configuration for production deployments.

8.1 Network Architecture

8.2 Admin API Authentication

8.3 Container App Secrets

8.4 VNet Integration

8.5 Non-Root Container

8.6 Secrets in Transformations

8.7 TLS / HTTPS

9 CI/CD Integration

Automate transformation deployment with Azure DevOps, GitHub Actions, or any pipeline that can call HTTP.

9.1 The Deployment Model

9.2 Bundle Repository Structure

9.3 GitHub Actions Workflow

9.4 Azure DevOps Pipeline

9.5 Rollback

9.6 Multi-Environment Promotion

9.7 Testing in the Pipeline

9.8 GitOps Pattern

10 Troubleshooting

Common problems, their symptoms, and how to fix them.

10.1 Container Starts But ready=false

10.2 Transformation Upload Fails (400)

10.3 Messages Failing (500 on Data Plane)

10.4 High Latency (p99 > 100ms)

10.5 Out of Memory (OOM Kill)

10.6 Admin API Returns 403

10.7 Transformations Lost After Restart

10.8 Dapr Not Delivering Messages

10.9 Schema Validation Failing Unexpectedly

10.10 Useful Diagnostic Commands

11 Appendix A: Admin API Reference

Complete reference for all Admin API endpoints on port 8081 and data plane endpoints on port 8085.

11.1 Authentication

11.2 Bundle Endpoints

11.3 Transformation Endpoints

11.4 Testing Endpoint

11.5 Operational Endpoints

11.6 Validation Override Endpoints

11.7 Schema Endpoints

11.8 Engine Endpoints

11.9 Health Endpoints (no authentication)

11.10 Data Plane Endpoints (port 8085, no authentication)

11.11 Response Codes

12 Appendix B: Configuration Reference

All configuration options: environment variables, engine.yaml, and transform.yaml.

12.1 Environment Variables**12.2 Engine Configuration (engine.yaml)****12.3 Transformation Configuration (transform.yaml)****12.4 JVM Tuning****12.5 Container App Plans****12.6 Bundle ZIP Format****12.7 Data Directory Layout****13 Appendix C: UTL-X Quick Reference**

Syntax cheat sheet for the most common transformation patterns. For the complete language reference, see “UTL-X: One Language, All Formats.”

13.1 File Structure**13.2 Format Headers****13.3 Property Access****13.4 Object Construction****13.5 Let Bindings****13.6 Conditionals****13.7 Array Operations****13.8 String Functions****13.9 Math Functions****13.10 Type Functions****13.11 Date Functions****13.12 Security Functions****13.13 User-Defined Functions****13.14 Pipe Operator**